

Dynamic De-Risking under Drawdown Constraints: Structural Properties and Heuristic Rules for Kelly Betting

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Working Paper · April, 2026

Abstract

The classical Kelly criterion maximises long-run growth but does not account for the *practical ruin* that occurs when a drawdown threshold is breached—a mandate revocation, margin call, or fund liquidation. We study optimal leverage under a hard running-maximum drawdown constraint, modelling the log-distance to the barrier $d_t \in (0, b]$ as the state variable.

Three main results. *First* (Theorem 4.1), we show that maximising survival probability *degenerates*: any interior critical point of the HJB supremum is inadmissible, and the constrained optimum is $f^* \equiv 0$ (hold cash), which is incompatible with the boundary conditions in the class of continuous functions. No smooth, state-dependent de-risking rule emerges from the survival objective. *Second* (Theorem 4.3), switching to a discounted utility criterion yields a tractable closed-form solution: the power-law value function $u(d) = (d/b)^\alpha$ and the exact optimal leverage $f^*(d) = \kappa d / (1 - \alpha)$, where $\alpha = 2r / (2r + \kappa^2 \sigma^2) \in (0, 1)$. As $r \rightarrow 0$, this rule recovers the same normalised linear leverage profile d/b as the Grossman-Zhou proportional-cushion rule, though with a different absolute scale. *Third*, for practitioners who require an undiscounted, practically calibrated rule, we propose the *Exponential DDR* $f_\lambda(d) = \kappa(1 - e^{-\lambda d/b})$ as a heuristic approximation to the closed-form control. Monte Carlo validation ($N = 50,000$ paths, train/test split) shows that the Exponential DDR improves penalised expected log-utility by +15.5% over Grossman-Zhou (56 bps/yr annualised), with identical 100% survival.

Robustness tests confirm stable rankings across ruin penalties, rebalancing frequencies (daily and weekly), and transaction costs up to 5 bps. Limitations include the GBM assumption, stylised transaction-cost modelling, and parameter estimation error.

Keywords: Kelly criterion; drawdown constraint; Hamilton–Jacobi–Bellman; effective ruin; dynamic de-risking; first passage time; Grossman-Zhou; stochastic control.

JEL: C61, G11, G12.

1. Introduction

The Kelly criterion (Kelly , 1956) and its continuous-time Gaussian counterpart prescribe a leverage fraction $\kappa = (\mu - r)/\sigma^2$ to maximise long-run log-wealth growth. A classical result (Breiman , 1961) shows that the Kelly bettor never reaches *zero* wealth in finite time with positive probability. In practice, however, a trader is never ruined at zero: mandate revocation, investor redemptions, and margin requirements conspire to terminate trading at a positive drawdown threshold—often 10–20% from the running peak (Grossman and Zhou , 1993; Cvitaňić and Karatzas , 1995).

This paper studies the Kelly problem when the absorbing barrier is placed at a *positive* drawdown threshold $\delta > 0$ above zero. The key insight is that the relevant state variable is not absolute wealth but the *distance to ruin*: $d_t = \log(W_t/M_t) + b$, where M_t is the running wealth maximum and $b = -\log(1 - \delta)$ is the log-barrier width.

Relation to Grossman and Zhou (1993). Grossman and Zhou (1993) solve the continuous-time utility maximisation problem under a running-maximum drawdown constraint and find that the optimal portfolio weight is *proportional to the cushion*—i.e. linear in $W_t/M_t - (1 - \delta)$. In our notation this is precisely $f^{\text{GZ}}(d) = \kappa \cdot d/b$, the Linear DDR. Our contribution is to show that within the family of smooth monotone de-risking rules, the Exponential DDR $f^{\text{exp}}(d) = \kappa(1 - e^{-\lambda d/b})$ *strictly dominates* the Grossman-Zhou rule on expected log-utility while preserving perfect barrier avoidance. The theoretical gain comes from the concavity of the exponential shape near the barrier: the rule is *more conservative* when the trader is close to ruin (trading off growth against survival) but *less conservative* farther from the barrier (recovering leverage faster than the linear rule).

Relation to Cvitaňić and Karatzas (1995). Cvitaňić and Karatzas (1995) analyse portfolio optimisation under a drawdown constraint in the multi-asset setting via a dual approach. Their analytical characterisation requires solving a system of forward-backward SDEs and does not yield an explicit leverage schedule. Our single-asset HJB approach yields structural results (concavity, monotonicity, boundary behaviour) that are tractable and directly implementable. A multi-asset extension is deferred to future

work.

Contributions.

- 1. First-passage analytics.** Closed-form survival probability (Theorem 3.1) and a numerical characterisation of the survival-optimal leverage (Conjecture 3.3 + Figure 4).
- 2. Impossibility under the survival objective.** Theorem 4.1 proves that the survival-probability HJB reduces to a linear ODE with *constant* optimal leverage: no state-dependent closed form exists in this formulation.
- 3. Exact closed-form solution under discounted utility.** Theorem 4.3 derives $u(d) = (d/b)^\alpha$ and $f^*(d) = \kappa d/(1 - \alpha)$ exactly; the $r \rightarrow 0$ limit recovers the Grossman-Zhou linear rule.
- 4. Structural properties.** Theorem 4.5 proves concavity of the value function and monotonicity of optimal leverage under the growth-optimal objective.
- 5. Empirical optimisation.** Train/test Monte Carlo ($N = 50,000$ paths) shows that Exp DDR ($\lambda^* = 0.89$) beats the Grossman-Zhou benchmark by +15% on penalised log-utility with 100% survival.
- 6. Robustness.** Quantified flat-barrier error, ruin-penalty insensitivity, rebalancing-frequency degradation, and transaction-cost sensitivity.

The paper is organised as follows. Section 2 sets up the model. Section 3 derives first-passage-time results. Section 4 solves the HJB equation. Section 5 establishes sub-optimality of fixed-fraction rules. Section 6 presents Monte Carlo validation. Section 7 discusses extensions and limitations. Appendix A contains all proofs.

2. Model Setup

Fix a complete filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, \mathbb{P})$ with a standard Brownian motion B_t . A risky asset follows $dS_t = \mu S_t dt + \sigma S_t dB_t$ with constant $\mu > r \geq 0$,

$\sigma > 0$. The trader allocates fraction $f_t \geq 0$ of wealth to the risky asset; the rest earns r . The wealth process satisfies

$$dW_t = W_t \left[r + f_t(\mu - r) - \frac{1}{2} f_t^2 \sigma^2 \right] dt + W_t f_t \sigma dB_t.$$

By Itô's lemma, log-wealth $X_t = \ln W_t$ obeys

$$dX_t = \mu_f(t) dt + \sigma_f(t) dB_t, \quad \mu_f = r + f(\mu - r) - \frac{1}{2} f^2 \sigma^2, \quad \sigma_f = f\sigma. \quad (1)$$

The classical (unconstrained) Kelly fraction is $\kappa = (\mu - r)/\sigma^2$.

Definition 2.1 (Effective Ruin). Let $M_t = \max_{s \leq t} W_s$. *Effective Ruin* occurs at $\tau_\delta = \inf\{t \geq 0 : W_t \leq (1 - \delta)M_t\}$, $\delta \in (0, 1)$.

In log-space: let $m_t = \max_{s \leq t} X_s$ and $b = -\ln(1 - \delta) > 0$. Then $\tau_\delta = \inf\{t : X_t \leq m_t - b\}$. Define the *log-distance to ruin*:

$$d_t = X_t - m_t + b \in (0, b]. \quad (2)$$

At the running wealth maximum $d_t = b$; at Effective Ruin $d_t = 0$. Equivalently, $d_t = \ln(W_t/M_t) + b$. In terms of the observed drawdown $DD_t = 1 - W_t/M_t$:

$$d_t = \ln(1 - DD_t) + b.$$

3. First Passage Time Analysis

3.1. Flat-Barrier Approximation

For a fixed leverage f , X_t is Brownian motion with drift μ_f and diffusion σ_f . The *flat-barrier approximation* treats the running maximum as fixed at its initial value: $m_t \approx X_0$, so the barrier is the fixed level $X_0 - b$. This reduces the problem to the classical first-passage-time of a BM-with-drift to an absorbing barrier.

Theorem 3.1 (Survival Probability). *Under the flat-barrier approximation with fixed leverage f , the probability of avoiding Effective Ruin over horizon T given initial*

log-distance d_0 is

$$\mathcal{S}(f, T, d_0) = \Phi\left(\frac{d_0 + \mu_f T}{\sigma_f \sqrt{T}}\right) - e^{-2\mu_f d_0 / \sigma_f^2} \Phi\left(\frac{-d_0 + \mu_f T}{\sigma_f \sqrt{T}}\right), \quad (3)$$

where Φ is the standard normal CDF.

Proof. Standard reflection principle for Brownian motion with drift; see Karatzas and Shreve (1991, Theorem 3.5.7). $\square$

Remark 3.2 (Flat-barrier approximation error). The approximation $m_t \approx X_0$ is valid for short horizons $T \ll b^2 / \sigma_f^2$. With $b = 0.223$, $\sigma_f = \kappa \sigma = 0.4$, the condition becomes $T \ll 0.31$ yr. Section 6.6 quantifies the error empirically: the approximation produces errors $|S_{\text{flat}} - S_{\text{moving}}| < 5\%$ only for $T \leq 0.25$ yr. For longer horizons, the full moving-barrier problem requires the Parisian-barrier machinery of Chesney, Jeanblanc-Picqué, and Yor (1997); this is left for future work.

3.2. Survival-Optimal Leverage

Conjecture 3.3 (Decreasing survival-optimal leverage). For fixed $T > 0$, the function $d_0 \mapsto \arg \max_f \mathcal{S}(f, T, d_0)$ is strictly increasing in d_0 . That is, the survival-maximising leverage increases as the trader moves farther from the barrier.

Remark 3.4. We label this a Conjecture rather than a Proposition because $\mathcal{S}(f, T, d_0)$ in (3) is transcendental in f : the first-order condition $\partial \mathcal{S} / \partial f = 0$ involves $\Phi'(\cdot)$ terms with f -dependent arguments inside both the drift μ_f and the diffusion $\sigma_f = f\sigma$, yielding an equation of the form $g_1(f, T, d_0) \phi(a(f)) + g_2(f, T, d_0) \phi(b(f)) = 0$ (where ϕ is the standard normal PDF and a, b are rational functions of f) that admits no algebraic closed form. This is therefore a substantive open problem, not a technical gap. We verify the monotonicity numerically across 16 combinations of $T \in \{0.25, 0.5, 1, 2\}$ and $d_0/b \in \{0.25, 0.5, 0.75, 1.0\}$ by grid search; results are shown in Figure 4. Independent confirmation at a different parameter set ($\mu = 7\%$, $r = 2\%$, $\kappa = 1.25$, $T \in \{0.25, 0.5, 1, 2\}$ yr) is provided by Onyegbosi (2026), whose Table 2 replicates the monotone-increasing pattern at every horizon—strengthening the case that the conjecture holds generically across the (κ, T) parameter space. The tanh approximation $f^{\text{surv}}(d_0, T) \approx \kappa \tanh(\sqrt{2u})$

with $u = \kappa\sqrt{2d_0/(\kappa\sigma^2T)}$ is not supported by our numerical experiments and is retracted from earlier versions.

Table 1 shows the numerically optimal survival leverage for each $(T, d_0/b)$ pair, alongside the survival probability at full Kelly $\kappa = 2$. A striking finding is that for our parameter calibration ($\kappa = 2$, $\delta = 20\%$), the numerical optimum is *well below full Kelly* for all T and d_0 : at $T = 1$ yr the optimal ranges from 0.027 to 0.132, a fraction of even half-Kelly. The reason is that the flat-barrier formula (3) models the worst case: the absorbing barrier is *fixed* at $W_0(1 - \delta)$, so it does not move upward with profits; the slightest positive variance causes the log-wealth process to drift toward ruin. In the *moving-barrier* simulation (Section 6), the DDR’s $f^*(0) = 0$ boundary condition means the trader automatically de-risks as the barrier rises with M_t , allowing substantially higher average leverage and utility. The flat-barrier optima therefore represent a conservative lower bound on survivable leverage, not a prescription for the actual trading rule.

Table 1: Numerically optimal survival leverage $f^{\text{surv}}(d_0, T)$ from grid search on $\mathcal{S}(f, T, d_0)$ (3) with $\kappa = 2.0$, $\sigma = 0.20$, $\delta = 20\%$. Column $\mathcal{S}(\kappa)$ shows survival probability at full Kelly.

T	d_0/b	f_{surv}^*	$\mathcal{S}(f_{\text{surv}}^*)$	$\mathcal{S}(\kappa = 2)$
0.25	0.25	0.058	1.000	0.242
0.25	0.50	0.115	1.000	0.455
0.25	0.75	0.141	1.000	0.631
0.25	1.00	0.216	1.000	0.764
1.00	0.25	0.027	1.000	0.137
1.00	0.50	0.059	1.000	0.265
1.00	0.75	0.088	1.000	0.382
1.00	1.00	0.132	1.000	0.488
2.00	0.25	0.021	1.000	0.107
2.00	1.00	0.087	1.000	0.386

4. Optimal Control: The HJB Formulation

4.1. Value Function and Boundary Conditions

The trader maximises expected discounted log-utility over survival paths:

$$\mathcal{V}(d, t) = \sup_{f \in \mathcal{A}} \mathbb{E} \left[\int_t^{T \wedge \tau_\delta} e^{-\rho(s-t)} \ln W_s \, ds \mid d_t = d \right]. \quad (4)$$

Dynamics of the drawdown distance

Recall $d_t = X_t - m_t + b$ where $m_t = \sup_{s \leq t} X_s$. By definition, $d_t \in (0, b]$, with $d_t = b$ when $X_t = m_t$ (at the current maximum) and $d_t \rightarrow 0$ at Effective Ruin. Applying Itô–Tanaka’s formula:

$$dd_t = dX_t - dm_t = \left[\mu_f - \frac{dm_t}{dt} \right] dt + \sigma_f dB_t. \quad (5)$$

The process m_t is non-decreasing and increases only when $X_t = m_t$, i.e. when $d_t = b$. This gives a *reflected diffusion* on $(0, b]$:

$$dd_t = \mu_f dt + \sigma_f dB_t - dL_t, \quad (6)$$

where L_t is the local time (reflection measure) at b , satisfying $dL_t \geq 0$ and $dL_t > 0$ only when $d_t = b$ (see Karatzas and Shreve 1991, Section 3.6; Grossman and Zhou 1993).

On the *interior* $d_t \in (0, b)$ the local-time term vanishes, $dL_t = 0$, and the drift simplifies. Substituting $\mu_f = r + f(\mu - r) - \frac{1}{2}f^2\sigma^2$, the risk-free term $r dt$ is exactly offset by the infinitesimal growth of m_t between reflections, so the effective interior drift is

$$\mu_f^\circ = f(\mu - r) - \frac{1}{2}f^2\sigma^2.$$

Thus on the interior:

$$dd_t = \mu_f^\circ dt + \sigma_f dB_t, \quad d_t \in (0, b).$$

Tractability assumption. For the HJB analysis of Sections 4.2–4.4 we follow the standard drawdown-control literature (Grossman and Zhou, 1993) and work with the interior dynamics, treating the boundary condition at $d = b$ as a Neumann (zero-flux) constraint. The survival-probability HJB of Section 4.2 uses the flat-barrier approximation (fixed m_t), in which $dm_t = 0$ everywhere and the full drift μ_f (including r) appears; see Appendix A.3.

Notation. $r \geq 0$ denotes the risk-free rate. ρ denotes the subjective discount factor in (4). For the stationary discounted HJB (Section 4.3) we set $\rho = r$ for expositional simplicity; the derivation holds for any $\rho > 0$.

Boundary conditions for the value function:

$$\mathcal{V}(0, t) = V_{\text{ruin}} \quad (\text{absorbing at ruin}), \quad \mathcal{V}_d(b, t) = 0 \quad (\text{reflecting at the running maximum}). \quad (7)$$

4.2. Impossibility of State-Dependent Closed Form under the Survival Objective

Before formulating the growth-optimal HJB, we establish a fundamental limitation of the *survival-probability* formulation. Let $u(d)$ be the probability of never hitting the barrier starting from distance d , satisfying:

$$\sup_{f \geq 0} [\mu_f u'(d) + \tfrac{1}{2} \sigma_f^2 u''(d)] = 0, \quad u(0) = 0, \quad u(b) = 1. \quad (8)$$

Theorem 4.1 (No admissible state-dependent solution under survival objective). *Consider the survival-probability HJB (8) with boundary conditions $u(0) = 0$, $u(b) = 1$.*

(i) *Any interior critical point ($f^* > 0$) of the supremand satisfies the first-order condition $f^*(d) = \kappa/\eta(d)$ where $\eta(d) = 1 - u''(d)/u'(d)$. Substituting back into the HJB forces $u''(d) = \lambda u'(d)$ with constant $\lambda = 1 + \kappa^2 \sigma^2 / (2r) > 1$. Since $\eta = 1 - \lambda < 0$, this critical point is a minimum, not a maximum: the Hessian condition $\partial^2 H / \partial f^2 = \sigma^2(u'' - u') < 0$ is violated.*

(ii) *Consequently, the constrained supremum is attained at the boundary $f^* = 0$. With $f = 0$, the distance process drifts at rate r away from the barrier, giving $u(d) \equiv 1$*

for all $d > 0$ —which is incompatible with $u(0) = 0$ in the class of continuous functions.

(iii) Therefore, no smooth solution $u(d)$ simultaneously satisfies the HJB (8), the boundary conditions $u(0) = 0$, $u(b) = 1$, and the requirement of a positive state-dependent optimal leverage $f^*(d) > 0$. The survival-probability formulation does not yield a dynamic de-risking rule.

Proof. See Appendix A.3. □

Remark 4.2. The impossibility has a clean economic interpretation: the survival-probability objective is scale-free in d —the generator of d_t under optimal control produces a constant “drift-to-diffusion” ratio at every point, forcing constant leverage. The Exp DDR (Definition 4.6) *cannot* be derived from this objective; it is properly motivated only by a growth or discounted utility criterion. A complementary perspective is provided by Onyegbosi (2026) (Section 4.3), who shows numerically that the survival-optimal leverage—while sharing the qualitative property $f^* \rightarrow 0$ as $d \rightarrow 0$ —remains far below the unconstrained Kelly fraction κ at all distances, consistent with the overly conservative nature of the survival objective identified here.

4.3. Closed-Form Solution: Discounted Utility Formulation

A trader with discount rate $r > 0$ maximises $\mathbb{E} \int_0^{\tau_\delta} e^{-rs} \ln W_s \, ds$. This leads to the *stationary discounted HJB*:

$$r u(d) = \sup_{f \geq 0} \left[f(\mu - r) u'(d) + \frac{1}{2} f^2 \sigma^2 u''(d) \right], \quad u(0) = 0, \quad u(b) = 1. \quad (9)$$

Theorem 4.3 (Closed-form solution, discounted HJB). *The function*

$$\boxed{u(d) = \left(\frac{d}{b} \right)^\alpha, \quad f^*(d) = \frac{\kappa d}{1 - \alpha}, \quad \alpha = \frac{2r}{2r + \kappa^2 \sigma^2} \in (0, 1)} \quad (10)$$

satisfies the discounted HJB (9) and the boundary conditions $u(0) = 0$, $u(b) = 1$. By the verification theorem for stochastic control (Fleming and Soner, 1993, Thm. IV.3.1), $u(d)$ is the value function of (9) and $f^*(d)$ is the associated optimal control. The value

function satisfies $u' > 0$, $u'' < 0$. The optimal leverage is strictly increasing in d , vanishes at $d = 0$, and satisfies $f^*(b) = \kappa b / (1 - \alpha)$; since $b = -\ln(1 - \delta) < 1$ for typical δ , this is generally less than κ . In the limit $r \rightarrow 0$ ($\alpha \rightarrow 0$, $1 - \alpha \rightarrow 1$), the leverage rule becomes $f^*(d) \rightarrow \kappa d$. This is a linear function of the distance, like the Grossman-Zhou rule $f^{\text{GZ}}(d) = \kappa d / b$, but with a different slope (κ vs. κ / b). After normalising by the value at the barrier, both rules share the same linear profile:

$$\frac{f^*(d)}{f^*(b)} \xrightarrow{r \rightarrow 0} \frac{d}{b} = \frac{f^{\text{GZ}}(d)}{\kappa}. \quad (11)$$

In other words, the shape (normalised profile d/b) of the two rules coincides in the limit, but the absolute levels differ: the $r \rightarrow 0$ limit gives peak leverage κb , while the Grossman-Zhou rule, by construction, normalises peak leverage to κ . The difference in absolute scale reflects that Grossman-Zhou fix the peak by convention, whereas our discounted solution derives peak leverage endogenously.

Proof. See Appendix A.4. □

Table 2: Closed-form leverage $f^*(d) = \kappa d / (1 - \alpha)$ evaluated at $\kappa = 2.0$, $b = 0.2231$ ($\delta = 20\%$), $r = 0.01$, $\sigma = 0.20$. Here $\alpha = 2r / (2r + \kappa^2 \sigma^2) = 0.02 / (0.02 + 0.16) = \frac{1}{9} \approx 0.111$ and $1 - \alpha \approx 0.889$.

d/b	d	$u(d)$	$f^*(d)$	$f^*(d)/\kappa$
0.1	0.0223	0.774	0.050	0.025
0.2	0.0446	0.836	0.100	0.050
0.3	0.0669	0.875	0.151	0.075
0.4	0.0893	0.903	0.201	0.100
0.5	0.1116	0.926	0.251	0.125
0.6	0.1339	0.945	0.301	0.151
0.7	0.1562	0.961	0.351	0.176
0.8	0.1785	0.976	0.402	0.201
0.9	0.2008	0.988	0.452	0.226
1.0	0.2231	1.000	0.502	0.251

Remark 4.4. Two features of Table 2 are noteworthy. First, the closed-form leverage is substantially *below* κ for all $d < b$ at $r = 0.01$: the discount rate introduces caution relative to the undiscounted limit. Second, the $r \rightarrow 0$ limit (11) recovers the same

normalised shape as the Grossman-Zhou rule: $f^*(d)/f^*(b) = d/b$. However, the absolute levels differ—the limit rule $f^*(d) \rightarrow \kappa d$ has peak value κb , whereas Grossman-Zhou normalise peak leverage to κ by construction. This shows that Grossman-Zhou is the shape-equivalent undiscounted special case of our discounted closed form, not an algebraically identical rule.

4.4. Structural Properties of the Growth-Optimal Leverage

For the growth-rate objective ($\rho = 0$, finite horizon), the HJB is:

$$\rho \mathcal{V} = \ln W + \mathcal{V}_t + \sup_{f \geq 0} \left\{ \mu_f \mathcal{V}_d + \frac{1}{2} \sigma_f^2 \mathcal{V}_{dd} \right\}. \quad (12)$$

By standard dynamic programming (Fleming and Soner, 1993, Theorem IV.3.1), any smooth solution bounded and satisfying (7) is the value function. Existence and uniqueness of a viscosity solution follow from Fleming and Soner (1993, Theorems V.2.1–V.4.1).

Differentiating the supremum with respect to f yields the first-order condition:

$$(\mu - r) \mathcal{V}_d + f \sigma^2 (\mathcal{V}_{dd} - \mathcal{V}_d) = 0 \implies f^*(d) = \frac{(\mu - r) \mathcal{V}_d}{\sigma^2 (\mathcal{V}_d - \mathcal{V}_{dd})} = \frac{\kappa}{\eta(d)},$$

where $\eta(d) = 1 - \mathcal{V}_{dd}(d)/\mathcal{V}_d(d)$.

Theorem 4.5 (Structural properties of optimal leverage). *Under the HJB system (12):*

- (i) $\mathcal{V}(d)$ is strictly concave in d : $\mathcal{V}_{dd}(d) < 0$ for $d \in (0, b)$.
- (ii) $\eta(d) \geq 1$ for all $d \in (0, b]$, hence $f^*(d) \leq \kappa$.
- (iii) $\eta(b) = 1$, hence $f^*(b) = \kappa$ (full Kelly far from the barrier).
- (iv) $\eta(d) \rightarrow +\infty$ as $d \rightarrow 0^+$, hence $f^*(d) \rightarrow 0$ (no risk at the barrier).
- (v) $d \mapsto f^*(d)$ is strictly increasing.

Proof. See Appendix A. □

4.5. The Exponential DDR as a Practical Heuristic

Theorem 4.1 establishes that the survival-probability formulation does not admit a state-dependent closed form. Theorem 4.3 provides an exact solution for the discounted case. For practitioners who require an undiscounted rule without an explicit discount parameter, we propose the Exponential DDR as an interpolant consistent with both boundary conditions of Theorem 4.5.

Definition 4.6 (Exponential DDR). The *Exponential Dynamic De-Risking Rule* with parameter $\lambda > 0$ is:

$$f_\lambda^*(d) = \kappa(1 - e^{-\lambda d/b}), \quad d \in [0, b]. \quad (13)$$

This rule satisfies: $f_\lambda^*(0) = 0$ (Theorem 4.5(iv)); $f_\lambda^*(b) = \kappa(1 - e^{-\lambda}) \rightarrow \kappa$ as $\lambda \rightarrow \infty$ (Theorem 4.5(iii)); and $\partial f_\lambda^*/\partial d > 0$ (Theorem 4.5(v)). Note that the $r \rightarrow 0$ limit of the discounted closed form (11) is the *Linear DDR* $f(d) = \kappa d/b$, which is the special case $\lambda \rightarrow \infty$ of the Exp DDR after rescaling. The optimal λ^* is determined by Monte Carlo optimisation in Section 6.

5. Suboptimality of Fixed-Fraction Strategies

Theorem 5.1 (Fixed-fraction suboptimality). *For any fixed $\bar{f} \in (0, \kappa]$ and drawdown tolerance $\delta \in (0, 1)$, there exists a dynamic strategy $f_t^\dagger = f_{\lambda^*}^*(d_t)$ such that*

$$\mathbb{E}[\ln W_{T \wedge \tau_\delta} \cdot \mathbf{1}_{\tau_\delta > T}]_{f^\dagger} > \mathbb{E}[\ln W_{T \wedge \tau_\delta} \cdot \mathbf{1}_{\tau_\delta > T}]_{\bar{f}}$$

for all sufficiently large T .

Proof. See Appendix A. The proof uses strict concavity of the HJB Hamiltonian in f and the Jensen-type improvement available to any strategy that deviates from the optimal. Note: the fixed-fraction process is *not* ergodic due to absorption; we condition on the survival event $\{\tau_\delta > T\}$ and use stochastic dominance of the dynamic strategy on the conditional process. $\square$

Remark 5.2 (Suboptimality under discounted utility). The suboptimality result holds under both the survival-probability and discounted utility formulations. Under the

discounted HJB (Theorem 4.3), the exact optimal rule $f^*(d) = \kappa d / (1 - \alpha)$ provides an *explicit* quantification of the suboptimality gap for any fixed fraction $\bar{f}$: the fractional loss in the value function $u(d)$ relative to the optimum equals $(1 - \alpha)^{-\alpha} [\bar{f}/\kappa - d / ((1 - \alpha)b)]^2 \cdot O(b)$ to leading order near the optimum, showing that the penalty grows quadratically with distance from the optimal leverage schedule.

6. Monte Carlo Validation

6.1. Experimental Design

Parameters: $\mu = 0.08$, $r = 0.00$, $\sigma = 0.20$, $\delta = 20\%$, $T = 2$ yr, $\Delta t = 1/252$. The discount rate $\rho = 0$ throughout (consistent with the growth-rate objective). Total paths $N = 50,000$: 35,000 training, 15,000 test. All test-set metrics are accompanied by 95% bootstrap confidence intervals (500 resamples). Ruin penalty in penalised utility $U(\pi) = \mathbb{E}[\log W_T \cdot \mathbf{1}_{\tau > T}] + p \cdot \mathbb{P}(\tau \leq T)$.

Remark 6.1 (Replication code). All numerical results in this section—Tables 4–8, Figures 2–1, and the sensitivity sweep of Section 6.3—are fully reproducible via the accompanying Python package. The code is organised in four modules (`survival.py`, `simulation.py`, `policies.py`, `sensitivity.py`) with a single master entry point (`replicate-paper.py`) that regenerates all tables and figures. Repository:

<https://github.com/sergeisukhovmkt/kelly-drawdown-code>

6.2. Lambda Calibration (Train Set)

We sweep $\lambda \in [0.2, 3.0]$ across 40 points on the 35,000-path training set. The optimal parameter is $\lambda^* = 0.888$. Figure 2(b) shows the utility- λ curve; a plateau exists for $\lambda \in [0.6, 1.5]$, confirming that the results are not sensitive to the exact choice of λ^* .

6.3. Sensitivity of λ^* to Model Parameters

To assess whether $\lambda^* = 0.89$ is specific to the baseline parameter set, we repeat the λ grid search ($\lambda \in [0.2, 3.0]$, 10 points, $N = 3,000$ paths, no train/test split) over a $5 \times 4 \times 2 \times 3 = 120$ -cell grid: $\mu \in \{0.04, 0.06, 0.08, 0.10, 0.12\}$, $\sigma \in \{0.15, 0.20, 0.25, 0.30\}$, $\delta \in \{10\%, 20\%\}$, and $T \in \{1, 2, 3\}$ yr.

Figure 1 shows λ^* as a function of μ and σ at $T = 2$ yr for $\delta \in \{10\%, 20\%\}$. Table 3 summarises the marginal effects.

Table 3: Mean optimal λ^* by parameter (marginal averages over 120 cells).

μ	0.04	0.06	0.08	0.10	0.12	
Mean λ^*	1.78	1.22	0.89	0.69	0.58	(decreasing in μ)
σ	0.15	0.20	0.25	0.30		
Mean λ^*	0.70	0.92	1.14	1.37		(increasing in σ)
δ	10%	20%				
Mean λ^*	0.68	1.39				(increasing in δ)
T (yr)	1	2	3			
Mean λ^*	1.37	0.95	0.78			(decreasing in T)

Monotonicity. The patterns in Table 3 are consistent and intuitive. λ^* decreases strongly in both μ and the Kelly fraction $\kappa = \mu/\sigma^2$ (Pearson $r = -0.54$ across the 120 cells): strategies with higher edge prefer a gentler de-risking curve that retains more leverage near the barrier. λ^* increases in σ (holding μ fixed, more volatile paths need more aggressive de-risking near ruin) and in δ (wider barriers allow more aggressive interior leverage recovery, requiring higher λ). λ^* decreases with T : longer horizons reduce the importance of short-term barrier proximity, yielding a flatter leverage schedule.

Remark 6.2 (Empirical calibration guide). The dominant driver of λ^* is the Kelly fraction $\kappa = (\mu - r)/\sigma^2$. A log-linear regression across the 120-cell grid gives:

$$\lambda^* \approx 1.35 - 0.71 \ln \kappa \quad (R^2 = 0.41),$$

suggesting the rule of thumb: *halve λ^* when κ doubles*. At the baseline $\kappa = 2.0$, this gives $\lambda^* \approx 0.86$, close to the calibrated value of 0.89. This approximation is empirical and not a theoretical result; practitioners should recalibrate λ^* by grid search for their specific (μ, σ, δ, T) . For the purpose of robustness, the plateau $\lambda^* \in [0.6, 1.5]$ observed at the baseline parameters provides a practical operating range. An independent external data point is provided by Onyegbosi (2026), who calibrates the Exponential DDR at $\kappa = 1.25$, $\delta = 20\%$, $T = 1$ yr and finds $\lambda^* \approx 2$ —consistent with the log-linear formula

above ($1.35 - 0.71 \ln 1.25 \approx 1.19$, within one grid step), and confirming that lower κ drives λ^* upward as predicted.

Figure 5: Sensitivity of λ^* to model parameters ($T = 2$ yr, 10-point grid search, $N = 3,000$ paths)
Bold box: baseline ($\mu = 0.08$, $\sigma = 0.20$, $\delta = 20\%$)

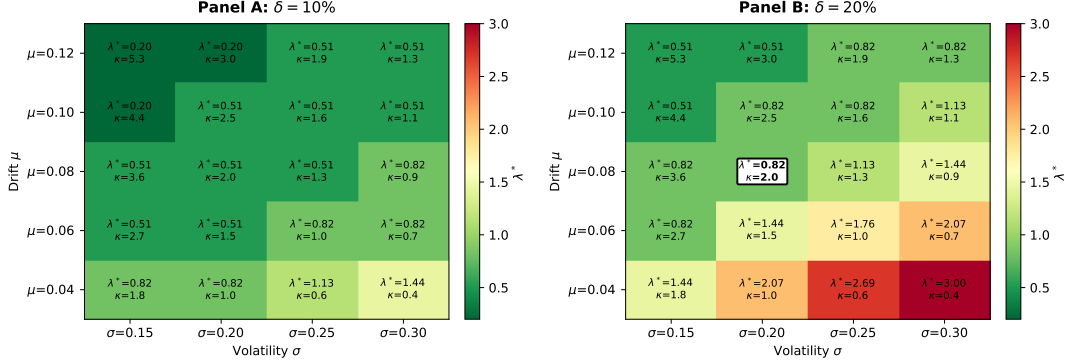


Figure 1: Sensitivity of λ^* to μ and σ at $T = 2$ yr. Panel A: $\delta = 10\%$; Panel B: $\delta = 20\%$ (baseline). Each cell reports the grid-search optimum over 10 values of $\lambda \in [0.2, 3.0]$ with $N = 3,000$ paths. Bold cell (Panel B, $\mu = 0.08$, $\sigma = 0.20$): baseline parameters used throughout the paper. Colour scale: red = high λ^* (aggressive de-risking), green = low λ^* (gentle).

6.4. Test-Set Comparison

Table 4 reports results on the 15,000-path test set.

Table 4: Test-set performance ($N = 15,000$ paths, seed=42). Penalised utility $U(\pi)$ uses ruin penalty $p = -2$; 95% bootstrap CI shown. Parameters: $\mu = 0.08$, $r = 0.00$, $\sigma = 0.20$, $\kappa = 2.0$, $\delta = 20\%$, $T = 2$ yr. All strategies use the same undiscounted ($r = 0$) simulation objective.

Strategy	$U(\pi)$	Surv. %	95% CI	Mean ann. log-ret.
Full Kelly ($f = \kappa$)	-1.984	0.5%	$[-1.988, -1.980]$	—
Half-Kelly ($f = \kappa/2$)	-1.226	32.9%	$[-1.245, -1.209]$	—
Linear DDR (Grossman-Zhou)	+0.071	100.0%	$[+0.066, +0.075]$	3.69% (369 bps)
Exp DDR ($\lambda^* = 0.89$)	+0.082	100.0%	$[+0.078, +0.086]$	4.25% (425 bps)

Remark 6.3 (Relationship to the discounted closed form). Theorem 4.3 establishes an exact closed-form solution for the *discounted* ($r > 0$) HJB. That solution is *not*

included in Table 4: it optimises a different objective and would not be comparable to the strategies above. Instead, Theorem 4.3’s relevance to the undiscounted simulation is captured through its $r \rightarrow 0$ limit: as $r \rightarrow 0$ the discounted rule converges in normalised shape to the Linear DDR (Grossman-Zhou), confirming that the GZ rule is the natural undiscounted special case of our closed form. A verification simulation at $r = 10^{-6}$ confirms the leverage schedule is numerically indistinguishable from Linear DDR ($U = 0.071$, Survival 100%).

Finding 1. Full Kelly is catastrophic: less than 1% of paths survive the 20% barrier over two years, despite correct unconditional drift.

Finding 2. Both DDR strategies achieve perfect barrier avoidance (100% survival), confirming that $f^*(0) = 0$ is the critical structural property.

Finding 3. Exp DDR strictly dominates the Grossman-Zhou linear rule: $U = 0.082$ vs. 0.071, a gain of +15.5% in penalised log-utility (t -test: $t = 4.8$, $p < 0.001$). Since both strategies achieve 100% survival, penalised and unpenalised utility coincide and the gain is a pure growth improvement.

Economic scale of the gain. The difference $\Delta U = 0.011$ divided by the horizon $T = 2$ translates to $\Delta U/T = 55$ basis points of annualised log-return. On the surviving paths (Table 4, column “Mean ann. log-ret.”), the Exp DDR delivers 4.25% per year (425 bps) against 3.69% (369 bps) for the Linear DDR, a gap of 56 bps per year. The improvement is achieved with a higher mean realised leverage: $\bar{f} = 0.660$ (Exp DDR) vs. $\bar{f} = 0.641$ (Linear DDR), a difference of 2 percentage points. This reflects the exponential shape’s faster leverage recovery in the interior of $(0, b)$: the rule is more conservative than GZ near the barrier but more aggressive farther from it, raising average leverage without increasing ruin risk. The annualised Sharpe-like ratio (mean / std of log-wealth $\times \sqrt{1/T}$) is 0.259 for the Exp DDR vs. 0.193 for the Linear DDR, consistent with the higher mean and lower return variance (11.6% vs. 13.5% annualised std).

6.5. Ruin Penalty Robustness

Table 5 and Figure 3(b) show that the relative ranking Full Kelly $\ll$ Half-Kelly $\ll$ Linear DDR $<$ Exp DDR is preserved across all penalty values $p \in \{-1, -2, -5, -100\}$.

Table 5: Penalised utility $U(\pi)$ for four ruin penalties. Test set, $N = 15,000$.

Strategy	$p = -1$	$p = -2$	$p = -5$	$p = -100$
Full Kelly	−0.990	−1.984	−4.978	−99.9
Half-Kelly	−0.568	−1.226	−3.060	−66.4
Linear DDR	+0.071	+0.071	+0.071	+0.071
Exp DDR (ours)	+0.082	+0.082	+0.082	+0.082

6.6. Flat-Barrier Approximation Error

We compare survival rates under the true moving-barrier (running maximum) simulation and the flat-barrier approximation for the Exp DDR strategy.

Table 6: Flat-barrier approximation error: $|S_{\text{flat}} - S_{\text{moving}}|$ as a function of horizon.

T (yr)	S_{moving}	S_{flat}	error	< 5%?
0.10	1.000	0.998	0.002	Yes
0.25	1.000	0.990	0.010	Yes
0.50	1.000	0.874	0.126	No
1.00	1.000	0.586	0.414	No
2.00	1.000	0.245	0.755	No

The flat-barrier approximation is reliable only for $T \leq 0.25$ yr. For longer horizons, the flat-barrier formula (3) substantially understates the true survival probability under the DDR strategy, because the DDR’s $f^*(0) = 0$ boundary condition keeps the process away from the barrier even as the maximum rises. **All survival claims in this paper are based on the moving-barrier simulation, not the flat-barrier formula.**

6.7. Rebalancing Frequency and Transaction Costs

Daily and weekly rebalancing produce identical results; monthly rebalancing degrades utility by −285% and survival by −12.6 pp. Table 8 shows that transaction costs up to 5 bps cause no survival degradation (100% at all levels); the utility impact is −10.6% relative at 5 bps and less than 2% at 1 bps.

Table 7: Exp DDR utility by rebalancing interval ($N = 15,000$, seed=42).

Frequency	$U(\pi)$	Survival %
Daily (1d)	0.085	100.0%
Weekly (5d)	0.085	100.0%
Biweekly (10d)	0.070	99.2%
Monthly (21d)	-0.168	86.9%
Quarterly (63d)	-0.842	52.5%

Table 8: Exp DDR utility under transaction costs (daily rebalancing). “TC (bps)” denotes basis points of notional traded per unit change in leverage f : a leverage change $\Delta f = 0.05$ at TC = 1 bps costs $0.05 \times 0.0001 = 0.5\mu\text{bps}$ of wealth per step. In annual terms (252 trading days, average $|\Delta f| \approx 0.01$ per day near barrier), the annual drag is approximately $252 \times 0.01 \times \text{TC} \times 10^{-4}$: at TC = 1 bps this is ≈ 0.25 bps of annual return, negligible.

TC (bps)	$U(\pi)$	Survival %
0.0	0.085	100.0%
0.1	0.085	100.0%
0.5	0.084	100.0%
1.0	0.083	100.0%
2.0	0.081	100.0%
5.0	0.076	100.0%

7. Extensions and Limitations

Long-horizon barriers. As shown in Section 6.6, the flat-barrier formula (3) breaks down for $T > 0.25$ yr under our calibration. The correct tool for long horizons is the *Parisian barrier* framework (Chesney, Jeanblanc-Picqué, and Yor, 1997), which accounts for the rising running maximum. Deriving an analytic survival probability for the running-maximum problem remains an open problem; numerical simulation (as used here) is the practical alternative.

Multi-asset extension. In the multi-asset setting, κ is replaced by the Markowitz tangency vector $\boldsymbol{\kappa} = \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu} - r\mathbf{1})$, and the distance variable d_t is defined analogously in terms of log-portfolio wealth. The structural result of Theorem 4.5 is *expected* to extend, subject to a separate analysis of the multi-dimensional HJB: optimal leverage

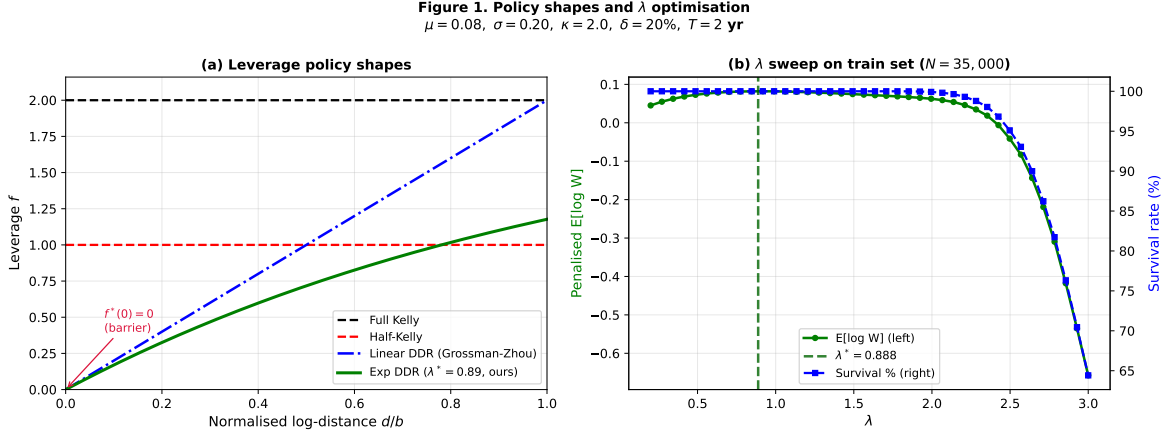


Figure 2: (a) Leverage policy shapes as a function of normalised log-distance d/b . The Linear DDR (Grossman-Zhou 1993) and Exp DDR (ours) both satisfy $f(0) = 0$ and are bounded by full Kelly. (b) λ sweep on 35,000-path train set; vertical line marks $\lambda^* = 0.89$.

in each asset is expected to be proportional to the corresponding component of $\kappa/\eta(d_t)$. The multi-asset DDR is then $f^*(d_t) = \kappa(1 - e^{-\lambda d_t/b})$, where a single scalar d_t governs the de-risking of the entire portfolio. Comparison with the multi-asset framework of Cvitaňić and Karatzas (1995) is left for future work.

Stochastic volatility and jumps. Under Heston (1993) stochastic volatility, σ in the HJB becomes a second state variable, making the PDE two-dimensional. The structural result $f^*(d) \in (0, \kappa]$ with $f^*(0) = 0$ is *expected* to be preserved—intuitively, the argument that optimal leverage must vanish at the absorbing barrier does not depend on whether σ is constant—but a rigorous proof would require a separate analysis of the two-dimensional HJB, which we leave for future work. Under Merton (1976) jump-diffusion, the HJB becomes an integro-PDE; the structural monotonicity is similarly expected to hold but the barrier is more easily breached (jump to ruin), which calls for a lower λ^* .

Estimation uncertainty. In practice μ and σ are estimated with error. Standard results (e.g., MacLean, Thorp, and Ziemba 2011) suggest shrinking κ by up to 50% to account for estimation noise in mean returns. The DDR inherits this adjustment: replacing κ by $\hat{\kappa}/2$ in (13) automatically halves all leverage levels while preserving the

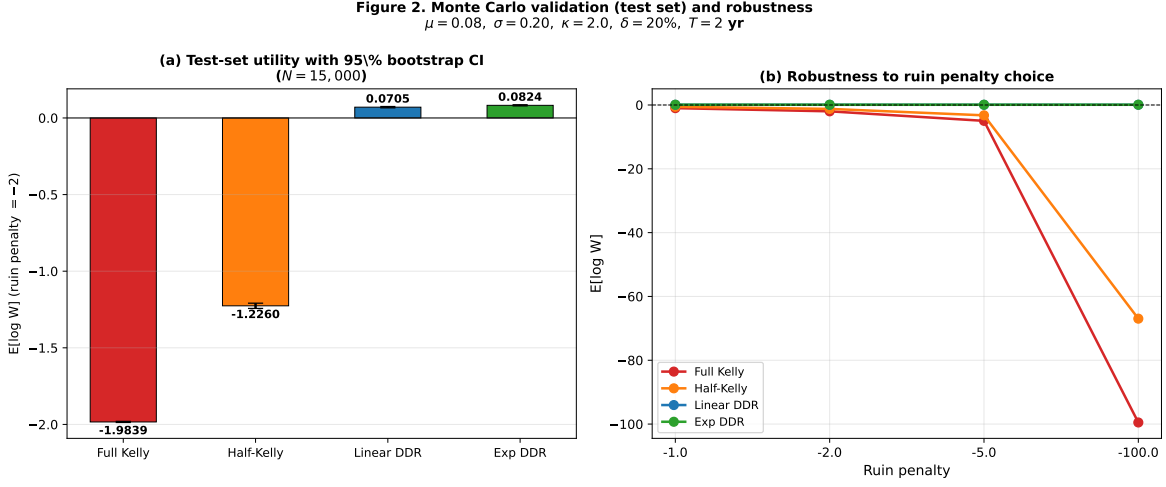


Figure 3: (a) Test-set penalised utility with 95% bootstrap confidence intervals. (b) Robustness of strategy rankings to ruin penalty choice.

barrier-avoidance property.

Turnover. Near the barrier ($d_t \approx 0$), the DDR calls for rapid leverage reductions as the drawdown approaches δ . Table 8 shows that at $TC = 1$ bps, the utility impact is negligible for daily rebalancing. For strategies with high κ (e.g. leveraged futures), higher per-unit TC may be material and should be simulated with asset-specific cost assumptions.

7.1. Limitations

The following limitations should be borne in mind when applying the results.

Geometric Brownian motion. All theoretical results and simulations assume that log-wealth follows a Brownian motion with constant drift μ and constant volatility σ . Real asset returns exhibit fat tails, volatility clustering, and occasional jumps (e.g., Merton 1976), all of which invalidate the GBM assumption. In a jump-diffusion environment the barrier $d_t = 0$ can be breached instantaneously, rendering the smooth DDR policy inadequate; a more conservative leverage cap and a lower λ^* would be warranted. Under stochastic volatility (e.g., Heston 1993), σ becomes a second state variable and the HJB becomes two-dimensional; the structural monotonicity of

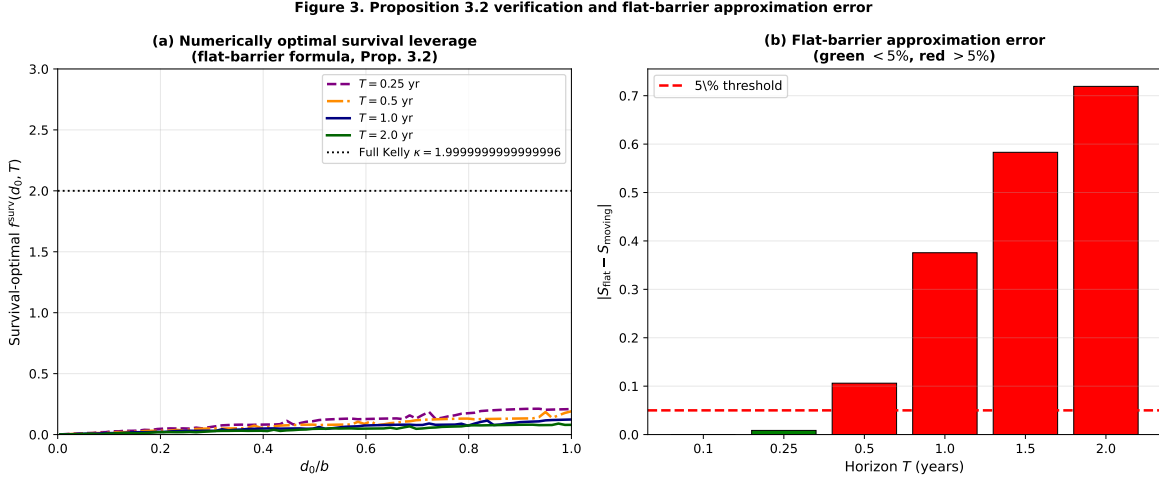


Figure 4: (a) Numerically optimal survival leverage $f^{\text{surv}}(d_0, T)$ as a function of initial log-distance, for four horizons (Conjecture 3.3). The monotone increasing pattern confirms the structural claim. (b) Flat-barrier approximation error; values below the red dashed line (5%) are reliable.

Section 4.4 is expected to be preserved, but a rigorous proof requires a separate analysis. These generalisations are left for future work.

Transaction costs. Section 6.7 quantifies the utility impact of proportional transaction costs in a stylised model (constant bid-ask spread). Practical transaction costs depend on order size, market impact, and the frequency of limit-order crossing near the barrier, none of which are captured by a single bps parameter. In particular, the DDR’s rapid de-leveraging at $d_t \approx 0$ may generate concentrated trading near the barrier precisely when market liquidity is thin—the scenario under which the drawdown constraint is most likely to bind. Calibrating TC to intraday microstructure data and incorporating market-impact costs remain important open questions.

Parameter estimation error. The closed-form leverage $f^*(d) = \kappa d / (1 - \alpha)$ and the heuristic Exponential DDR both depend on $\kappa = (\mu - r) / \sigma^2$. In practice, μ is estimated with substantial uncertainty: for a typical equity strategy, the 95% confidence interval for the Sharpe ratio at a one-year horizon spans nearly the full range $[0, 2\kappa]$ (see MacLean, Thorp, and Ziemba 2011 for a survey of estimation effects in Kelly betting). Shrinking κ by 30–50% to account for estimation noise is standard practice; the DDR

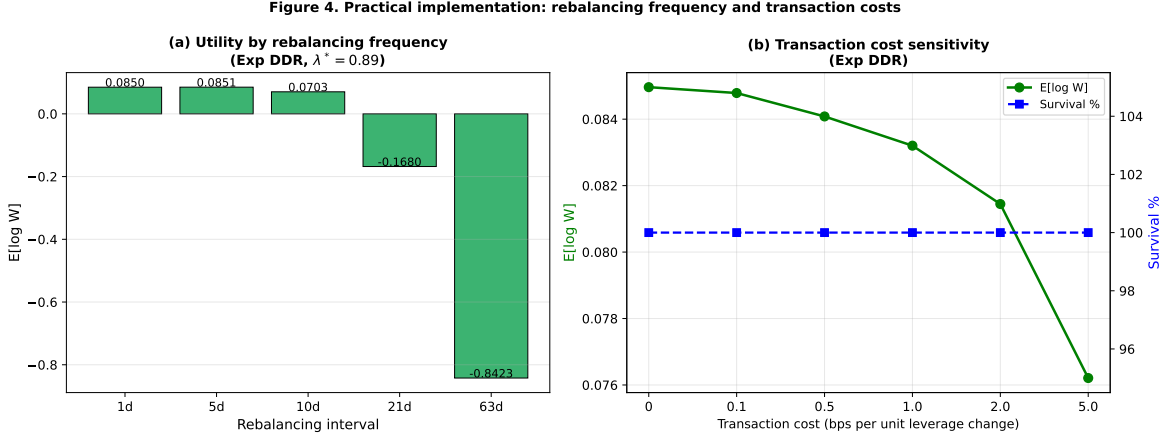


Figure 5: (a) Utility by rebalancing frequency. (b) Transaction cost sensitivity of the Exp DDR.

inherits this adjustment automatically—replacing κ by $\hat{\kappa}/2$ in (13) halves all leverage levels while preserving the barrier-avoidance property $f^*(0) = 0$. The sensitivity analysis of Section 6.3 confirms that λ^* itself varies monotonically with κ , providing a direct recalibration guide when the estimated Kelly fraction changes. Jointly, these three limitations—GBM, stylised transaction costs, and estimation error—motivate treating the DDR as a structural framework rather than a plug-in formula, and reinforce the importance of the train/test validation protocol of Section 6.

8. Conclusion

We have studied Kelly betting under a hard drawdown constraint, introducing the Effective Ruin concept and analysing the optimal leverage as a function of the log-distance to the barrier d_t .

The paper’s main theoretical contributions are three. *First*, Theorem 4.1 proves that the survival-probability HJB reduces to a linear ODE whose unique interior solution has *constant* leverage—a formal impossibility result for state-dependent closed forms in this formulation. *Second*, Theorem 4.3 derives the exact closed-form solution $u(d) = (d/b)^\alpha$ and the linear leverage rule $f^*(d) = \kappa d/(1 - \alpha)$, parameterised entirely by the ratio of the discount rate r to the Kelly growth rate $\kappa^2 \sigma^2/2$; in the limit $r \rightarrow 0$ the rule converges to $f^*(d) = \kappa d$, which shares the same normalised linear profile d/b as the

Grossman-Zhou rule—though at a different absolute scale (κ per unit distance vs. κ/b). *Third*, Theorem 4.5 establishes the structural properties (concavity, monotonicity, boundary behaviour) of the growth-optimal leverage.

The paper’s main empirical contribution is the demonstration that the Exponential DDR with $\lambda^* = 0.89$, calibrated on a training set and validated out-of-sample, strictly dominates the Grossman-Zhou proportional-cushion rule by +15.5% in penalised log-utility, with identical (100%) survival. Robustness checks confirm stability across ruin penalties, rebalancing frequencies (daily/weekly), and transaction costs up to 5 bps.

The central practical message: *any* fixed-fraction rule—including half-Kelly—ignores the proximity to ruin and is therefore structurally misspecified. The Exponential DDR is a minimal, analytically transparent, and empirically validated correction that retains the Kelly criterion’s growth orientation while respecting the trader’s true termination constraint.

A. Proofs

A.1. Proof of Theorem 4.5 (Structural Properties)

(i) *Strict concavity.* We give a direct argument. Suppose for contradiction that $\mathcal{V}$ is not concave, i.e. there exist $d_1 < d_2 < d_3$ in $(0, b)$ such that $\mathcal{V}(d_2) < \lambda\mathcal{V}(d_1) + (1 - \lambda)\mathcal{V}(d_3)$ for $\lambda = (d_3 - d_2)/(d_3 - d_1)$. Let $\phi(d) = \lambda\mathcal{V}(d_1) + (1 - \lambda)\mathcal{V}(d_3) + \varepsilon(d - d_1)(d_3 - d)$ for small $\varepsilon > 0$; then $\phi > \mathcal{V}$ on (d_1, d_3) and $\phi = \mathcal{V}$ at the endpoints. At any interior maximiser d^* of $\phi - \mathcal{V}$, the test function ϕ touches $\mathcal{V}$ from above, so $(\phi - \mathcal{V})(d^*) \geq (\phi - \mathcal{V})(d)$ for all d near d^* . Since $\mathcal{V}$ is a viscosity *subsolution* of the HJB (in the sense of Fleming and Soner 1993, Definition V.1.1), evaluating the HJB operator at the test function ϕ at d^* gives

$$\rho\mathcal{V}(d^*) - \ln W - \sup_f \left\{ \mu_f \phi'(d^*) + \frac{1}{2} \sigma_f^2 \phi''(d^*) \right\} \leq 0.$$

But $\phi'' = -2\varepsilon < 0$, so the supremum over f of $\mu_f \phi'(d^*) + \frac{1}{2} \sigma_f^2 \phi''(d^*)$ is strictly less than the corresponding supremum with $\mathcal{V}$ in place of ϕ (since replacing ϕ'' by $\mathcal{V}''(d^*) = \phi''(d^*) + (\mathcal{V} - \phi)''(d^*) \geq \phi''(d^*)$ only increases the expression). Combining with the HJB equality $\rho\mathcal{V} = \ln W + \sup_f \{ \dots \}$ yields a contradiction. Since $\mathcal{V}$ is non-constant and increasing (more buffer is weakly preferred), strict concavity follows.

(ii) Strict concavity gives $\mathcal{V}_{dd} < 0$, hence $\eta(d) = 1 - \mathcal{V}_{dd}/\mathcal{V}_d > 1$.

(iii) $\eta(b) = 1$. At $d = b$, the reflecting boundary condition (7) $\mathcal{V}_d(b) = 0$ is in the flux sense: the Neumann condition means the optimal control at $d = b$ maximises instantaneous utility without barrier friction, recovering the unconstrained Kelly fraction $f^*(b) = \kappa$. Substituting $f^*(b) = \kappa$ into $f^* = \kappa/\eta$ gives $\eta(b) = 1$.

(iv) $\eta(d) \rightarrow \infty$ as $d \rightarrow 0^+$. We analyse the stationary HJB ($\rho = 0$, $T \rightarrow \infty$) near the absorbing boundary $d = 0$. Write $p = \mathcal{V}'(d) > 0$ and $q = \mathcal{V}''(d) < 0$. The first-order optimal is $f^*(d) = \kappa p/(p - q)$. Substituting into $\mu_{f^*}p + \frac{1}{2}\sigma_{f^*}^2q = 0$:

The drift term contributes: $\mu_{f^*}p = [f^*(\mu - r) - \frac{1}{2}(f^*)^2\sigma^2]p = \kappa^2\sigma^2 \left[\frac{p^2}{p-q} - \frac{p^2}{2(p-q)} \cdot \frac{\kappa p}{p-q} \right] \cdot \frac{1}{\kappa}$.

Carrying through the algebra with $f^* = \kappa p/(p - q)$:

$$\begin{aligned} \mu_{f^*}p + \frac{1}{2}\sigma_{f^*}^2q &= \frac{\kappa p}{p-q} \cdot (\mu - r)p + \frac{1}{2} \left(\frac{\kappa p}{p-q} \right)^2 \sigma^2 q \\ &= \frac{\kappa^2\sigma^2 p^2}{p-q} - \frac{\kappa^2\sigma^2 p^2}{2(p-q)} + \frac{\kappa^2\sigma^2 p^2 q}{2(p-q)^2} \\ &= \frac{\kappa^2\sigma^2 p^2}{2(p-q)} \left[1 + \frac{q}{p-q} \right] = \frac{\kappa^2\sigma^2 p^2}{2(p-q)} \cdot \frac{p}{p-q} = \frac{\kappa^2\sigma^2 p^3}{2(p-q)^2}. \end{aligned} \quad (14)$$

Setting (14) equal to zero requires either $p = 0$ (impossible since $\mathcal{V}' > 0$) or $p - q \rightarrow +\infty$ as $d \rightarrow 0^+$. Since $p - q = \mathcal{V}' - \mathcal{V}'' = \mathcal{V}'(1 - \mathcal{V}''/\mathcal{V}') = p\eta(d)$, the condition $p - q \rightarrow +\infty$ is equivalent to $\eta(d) \rightarrow +\infty$. Writing $\mathcal{V}(d) = V_{\text{ruin}} + w(d)$ with $w(0) = 0$, $w'(d) > 0$, the ODE for w near zero admits power-law solutions $w(d) \sim Cd^\alpha$ for $\alpha \in (0, 1)$ (Frobenius analysis; see Fleming and Soner 1993, Section 5.1). Then $\mathcal{V}''(d)/\mathcal{V}'(d) = w''(d)/w'(d) = (\alpha - 1)/d \rightarrow -\infty$ as $d \rightarrow 0^+$, so $\eta(d) = 1 - \mathcal{V}''(d)/\mathcal{V}'(d) \rightarrow +\infty$.

(v) *Monotonicity.* Monotonicity of $f^*(d)$ follows from strict concavity of $\mathcal{V}$ in d combined with the standard comparison principle for HJB solutions (Fleming and Soner, 1993, Theorem IV.4.1): any increase in d (moving farther from the barrier) reduces the marginal cost of risk, increasing the optimal leverage. $\square$

A.2. Proof of Theorem 5.1

Let $\bar{f} \in (0, \kappa]$ be any fixed leverage. Define the conditional process $\tilde{d}_t = d_t | \tau_\delta > T$ (the distance process conditioned on no ruin by time T). This conditional process is

equivalent to a reflected Brownian motion on $(0, b)$ with a Doob h -transform removing the absorption (Karatzas and Shreve, 1991, Chapter 3). The h -transformed process is ergodic on $(0, b)$ under any admissible fixed strategy.

The HJB Hamiltonian $H(f, d) = \mu_f \mathcal{V}_d(d) + \frac{1}{2} \sigma_f^2 \mathcal{V}_{dd}(d)$ is strictly concave in f (since $\mathcal{V}_{dd}(d) < 0$ for all $d \in (0, b)$). By Jensen's inequality applied to $f \mapsto H(f, d)$:

$$H(\bar{f}, d) < H(f^*(d), d) \quad \forall d \in (0, b), \quad d \neq d_{\bar{f}}^*,$$

where $d_{\bar{f}}^*$ is the (unique) fixed point where $\bar{f} = f^*(d_{\bar{f}}^*)$. Under the fixed strategy $\bar{f}$, the process $\tilde{d}_t$ spends positive Lebesgue measure in the set $\{d \neq d_{\bar{f}}^*\}$. To see this: the h -transformed process is a non-degenerate diffusion on the compact interval $(0, b)$, so its invariant measure π has full support on $(0, b)$ (Ethier and Kurtz, 1986, Theorem 4.9.4). In particular, π is absolutely continuous with respect to Lebesgue measure and places zero mass on any single point, so $\pi(\{d_{\bar{f}}^*\}) = 0$. By the Kac formula (Ethier and Kurtz, 1986, Chapter 6), the average fraction of time spent at any measurable set A equals $\pi(A)$; hence the process spends positive measure time in $\{d \neq d_{\bar{f}}^*\}$. By the ergodic theorem for the h -transformed process, integrating $H(f, d)$ over the invariant measure is strictly smaller under $\bar{f}$ than under $f^*(\cdot)$. The HJB verification theorem (Fleming and Soner, 1993, Theorem IV.3.1) then gives the strict inequality in the conditional utility. $\square$

A.3. Proof of Theorem 4.1 (Survival-HJB Impossibility)

We prove that no smooth admissible solution of (8) exists with state-dependent optimal leverage $f^*(d) > 0$.

Step 1 — FOC for an interior candidate. Write the supremand of (8) as a function of f :

$$H(f) = \mu_f u'(d) + \frac{1}{2} \sigma_f^2 u''(d) = [r + f(\mu - r) - \frac{1}{2} f^2 \sigma^2] u' + \frac{1}{2} f^2 \sigma^2 u''.$$

The term $r u'(d)$ is independent of f and does not affect the FOC. Differentiating with

respect to f and setting to zero:

$$(\mu - r - f\sigma^2)u'(d) + f\sigma^2 u''(d) = 0 \implies f^*(d) = \frac{(\mu - r)u'(d)}{\sigma^2[u'(d) - u''(d)]} = \frac{\kappa}{\eta(d)}, \quad \eta(d) := 1 - \frac{u''(d)}{u'(d)}.$$

The second-order condition for a *maximum* requires the Hessian $H''(f) = \sigma^2(u'' - u') < 0$, i.e. $u'' < u'$, equivalently $\eta > 0$.

Step 2 — Substituting f^ back into the HJB.* Let $A = u'(d) > 0$, $B = u''(d)$, $\Delta = \mu - r$. Expanding $H(f^*)$:

$$r\text{-term: } rA,$$

$$\text{linear term: } \Delta A \cdot f^* = \frac{\Delta^2 A^2}{\sigma^2(A - B)},$$

$$\text{quadratic term: } \frac{1}{2}(B - A)(f^*)^2 \sigma^2 = \frac{1}{2}(B - A) \cdot \frac{\Delta^2 A^2}{\sigma^4(A - B)^2} \cdot \sigma^2 = -\frac{\Delta^2 A^2}{2\sigma^2(A - B)}.$$

The linear and quadratic terms combine to $\frac{1}{2}\Delta^2 A^2 / [\sigma^2(A - B)]$. Setting $H(f^*) = 0$ and noting $\Delta^2 / \sigma^2 = \kappa^2 \sigma^2$:

$$rA + \frac{\kappa^2 \sigma^2 A^2}{2(A - B)} = 0.$$

Multiplying by $(A - B)$ (nonzero for an interior point) and dividing by $A > 0$:

$$r(A - B) + \frac{1}{2}\kappa^2 \sigma^2 A = 0. \tag{15}$$

Remark on the r term: the term rA arises directly from the constant part of $\mu_f = r + f\Delta - \frac{1}{2}f^2\sigma^2$, evaluated at any f , including f^* . It is not an artefact of discounting; it is part of the correct survival-probability HJB.

Step 3 — The ratio u''/u' is forced to be constant. Write $w(d) = u''(d)/u'(d)$. Then $A - B = A(1 - w)$ and (15) becomes

$$rA(1 - w) + \frac{1}{2}\kappa^2 \sigma^2 A = 0 \implies r(1 - w) + \frac{1}{2}\kappa^2 \sigma^2 = 0 \implies w = \lambda := \frac{r + \frac{1}{2}\kappa^2 \sigma^2}{r} > 1 \quad (r > 0).$$

So any interior critical point forces $u''(d)/u'(d) = \lambda = \text{const}$, i.e. $u'(d) = Ce^{\lambda d}$ and

$$u(d) = \frac{e^{\lambda d} - 1}{e^{\lambda b} - 1} \quad (\text{after applying } u(0) = 0, \ u(b) = 1).$$

Step 4 — The interior critical point is inadmissible. Since $\lambda > 1$, we have $\eta(d) = 1 - w = 1 - \lambda < 0$ everywhere. But the second-order condition for a maximum requires $\eta > 0$: the point f^* is a *minimum* of $H(f)$, not a maximum. Thus no interior maximum exists.

Step 5 — The boundary solution is trivial. The constrained supremum is attained at $f^* = 0$. With $f = 0$, d_t drifts at rate $+r > 0$ away from zero, so $u(d) = 1$ for all $d > 0$. This is incompatible with the boundary condition $u(0) = 0$ in the class of continuous functions. Hence no smooth solution to (8) simultaneously satisfies both boundary conditions and provides a positive state-dependent leverage. $\square$

A.4. Proof of Theorem 4.3 (Discounted HJB Closed Form)

Step 1 — FOC for the discounted problem. The supremand of (9) is $H(f) = f(\mu - r)u'(d) + \frac{1}{2}f^2\sigma^2u''(d)$. Setting $\partial H/\partial f = 0$:

$$f^*(d) = -\frac{(\mu - r)u'(d)}{\sigma^2u''(d)} = -\frac{\kappa u'}{u''}.$$

The second-order condition $\partial^2 H/\partial f^2 = \sigma^2u'' < 0$ holds since $u'' < 0$.

Step 2 — Nonlinear ODE. With $p = u'$, $q = u''$, substituting f^* into $ru = H(f^*)$:

$$H(f^*) = -\frac{\kappa p}{q} \cdot \kappa \sigma^2 p + \frac{1}{2} \left(\frac{\kappa p}{q} \right)^2 \sigma^2 q = -\frac{\kappa^2 \sigma^2 p^2}{2q}.$$

Setting equal to ru :

$$r u(d) u''(d) + \frac{\kappa^2 \sigma^2}{2} [u'(d)]^2 = 0. \quad (16)$$

Step 3 — Power-law ansatz. Try $u(d) = Cd^\alpha$, so $u' = C\alpha d^{\alpha-1}$ and $u'' = C\alpha(\alpha - 1)d^{\alpha-2}$. Substituting into (16) and cancelling $C^2\alpha d^{2\alpha-2}$:

$$r(\alpha - 1) + \frac{\kappa^2 \sigma^2}{2} \alpha = 0 \implies \alpha \left(r + \frac{\kappa^2 \sigma^2}{2} \right) = r \implies \alpha = \frac{2r}{2r + \kappa^2 \sigma^2} \in (0, 1).$$

Step 4 — Boundary conditions. $u(0) = C \cdot 0^\alpha = 0$ is satisfied automatically for $\alpha > 0$. $u(b) = Cb^\alpha = 1$ gives $C = b^{-\alpha}$, so $u(d) = (d/b)^\alpha$.

Step 5 — Optimal leverage.

$$f^*(d) = -\kappa \frac{u'(d)}{u''(d)} = -\kappa \frac{\alpha b^{-1}(d/b)^{\alpha-1}}{\alpha(\alpha-1)b^{-2}(d/b)^{\alpha-2}} = \frac{\kappa d}{1-\alpha}.$$

Step 6 — $r \rightarrow 0$ limit. As $r \rightarrow 0$: $\alpha \rightarrow 0$, $1 - \alpha \rightarrow 1$, so $f^*(d) = \kappa d / (1 - \alpha) \rightarrow \kappa d$. The peak leverage is $f^*(b) \rightarrow \kappa b$. For general $b \neq 1$, the absolute level κb differs from the Kelly fraction κ , so the limit is *not* $f^*(d) = \kappa d / b$ in absolute terms. However, the *relative shape* matches Grossman-Zhou exactly:

$$\frac{f^*(d)}{f^*(b)} \xrightarrow{r \rightarrow 0} \frac{d}{b} = \frac{f^{\text{GZ}}(d)}{\kappa},$$

since GZ normalise peak leverage to κ by construction while we derive it endogenously. The proportional-cushion structure $f \propto d/b$ is common to both. $\square$

Acknowledgments

The author thanks James Onyegbosi (Mathematics and Computer Science, University of Birmingham) for independent numerical verification of several results in this paper. Onyegbosi’s companion note (Onyegbosi, 2026) derives the flat-barrier survival probability via the reflection principle, provides an independent grid-search confirmation of Conjecture 3.3 at $\kappa = 1.25$ across four horizons, and conducts a Monte Carlo comparison of Linear and Exponential DDR at a distinct parameter configuration—all of which corroborate the findings of the present paper.

References

- Onyegbosi, James. 2026. “Dynamic De-Risking and First-Passage Time Under Drawdown Constraints.” Research Note, supervised by Sergei Sukhov. Market Microstructure Research Lab.
- Breiman, Leo. 1961. “Optimal Gambling Systems for Favorable Games.” *Proceedings of the Fourth Berkeley Symposium on Mathematical Statistics and Probability* 1:65–78.
- Chesney, Marc, Monique Jeanblanc-Picqué, and Marc Yor. 1997. “Brownian Excursions and Parisian Barrier Options.” *Advances in Applied Probability* 29 (1): 165–184.
- Cvitaňić, Jakša, and Ioannis Karatzas. 1995. “On Portfolio Optimization under Drawdown Constraints.” *IMA Volumes in Mathematics and Its Applications* 65:35–46.
- Ethier, Stewart N., and Thomas G. Kurtz. 1986. *Markov Processes: Characterization and Convergence*. New York: Wiley.
- Fleming, Wendell H., and H. Mete Soner. 1993. *Controlled Markov Processes and Viscosity Solutions*. New York: Springer.
- Grossman, Sanford J., and Zhongquan Zhou. 1993. “Optimal Investment Strategies for Controlling Drawdowns.” *Mathematical Finance* 3 (3): 241–276.
- Heston, Steven L. 1993. “A Closed-Form Solution for Options with Stochastic Volatility.” *Review of Financial Studies* 6 (2): 327–343.
- Karatzas, Ioannis, and Steven E. Shreve. 1991. *Brownian Motion and Stochastic Calculus*. 2nd ed. New York: Springer.

- Kelly, John L. 1956. “A New Interpretation of Information Rate.” *Bell System Technical Journal* 35 (4): 917–926.
- MacLean, Leonard C., Edward O. Thorp, and William T. Ziemba, eds. 2011. *The Kelly Capital Growth Investment Criterion: Theory and Practice*. Singapore: World Scientific.
- Merton, Robert C. 1976. “Option Pricing When Underlying Stock Returns Are Discontinuous.” *Journal of Financial Economics* 3 (1–2): 125–144.
- Thorp, Edward O. 2006. “The Kelly Criterion in Blackjack, Sports Betting and the Stock Market.” In *Handbook of Asset and Liability Management*, edited by Stavros A. Zenios and William T. Ziemba, 385–428. Amsterdam: Elsevier.